

How Industrial Curing Methods Affect the Mechanical Properties of Additively Manufactured Dental Polymers

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Abstract. The rapid integration of additive manufacturing into dental prosthetics has made the post-processing of photopolymer resins a critical engineering bottleneck, as the final mechanical integrity of printed parts is heavily dictated by the curing phase, which often lacks industrial standardization in clinical workflows. This study investigates the impact of three distinct industrial curing protocols—conventional ambient-air ultraviolet (UV) curing, industrial nitrogen-inerted UV curing, and a hybrid thermal-UV curing process—on the mechanical properties and polymerization kinetics of two commercially available dental resins (a high-precision model resin and a load-bearing permanent crown resin). Standardized test specimens were fabricated using a high-resolution digital light processing (DLP) printer, uniformly washed in isopropanol, and subjected to the respective post-processing workflows. The empirical results demonstrated that industrial nitrogen-inerted UV curing significantly outperformed conventional ambient methods, yielding a 28% increase in flexural strength (from 85.4 ± 4.2 MPa to 109.3 ± 5.1 MPa) and a 22% improvement in surface microhardness, primarily by eliminating oxygen inhibition at the material surface during photopolymerization. Conversely, while the hybrid thermal-UV method achieved the highest overall degree of conversion (88.5%), it induced a 12% reduction in impact resistance due to excessive cross-linking density and subsequent polymer embrittlement. Statistical analysis via two-way ANOVA

1 Introduction

The integration of additive manufacturing (AM) technologies into dental prosthetics has undergone a paradigm shift over the past decade, evolving from rapid prototyping of diagnostic models to the direct fabrication of functional, load-bearing clinical restorations. Digital light processing (DLP) and stereolithography (SLA) systems now enable the production of surgical

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guides, temporary crowns, definitive prostheses, and even denture bases with sub-50-micron resolution and exceptional geometric fidelity. This technological maturation has been driven by the development of advanced photopolymer resins with biocompatibility certifications, improved mechanical performance, and aesthetic properties comparable to conventional subtractively manufactured materials. However, as dental additive manufacturing transitions from low-volume clinical customization to high-throughput industrial production, the engineering challenges associated with post-processing have emerged as a critical bottleneck that directly determines the final quality, reliability, and clinical longevity of printed components.

Unlike subtractive manufacturing processes such as milling, where the mechanical properties of the final part are largely inherited from the pre-polymerized blank material, additively manufactured photopolymers exist in a metastable, under-cured state immediately after the printing cycle. The layer-by-layer nature of DLP and SLA processes, combined with light scattering effects and oxygen diffusion at the resin-air interface, results in a polymer network with incomplete monomer conversion, residual stresses, and heterogeneous cross-linking density throughout the part volume. Consequently, the post-processing stage—comprising washing, secondary curing, and surface finishing—becomes the decisive engineering operation that transforms a fragile green-state prototype into a structurally reliable functional component. The curing phase, in particular, governs the final degree of conversion (DC), which directly correlates with flexural strength, microhardness, wear resistance, and biocompatibility of the polymer network.

Despite the critical importance of post-processing, industrial curing methodologies in dental laboratories and manufacturing facilities remain largely empirical and non-standardized. Conventional approaches rely on ambient-air ultraviolet (UV) curing chambers, which are simple and cost-effective but fundamentally limited by oxygen inhibition—a phenomenon where atmospheric oxygen scavenges free radicals at the material surface, preventing complete polymerization and resulting in a tacky, under-cured surface layer with compromised mechanical properties. This oxygen-inhibited layer not only reduces surface hardness and wear resistance but also serves as a potential source of unreacted monomer leaching, raising biocompatibility concerns. While some manufacturers recommend extended curing times or manual rotation of parts to mitigate these effects, such practices are labor-intensive, inconsistent, and incompatible with scalable industrial workflows.

Recent advances in industrial polymer processing have demonstrated that controlled-atmosphere curing, particularly nitrogen inerting, can effectively eliminate oxygen inhibition by displacing atmospheric oxygen from the curing chamber, thereby enabling complete surface polymerization and significantly enhancing mechanical performance. Similarly, hybrid thermal-UV curing protocols, which combine photopolymerization with controlled thermal energy to activate latent initiators and promote post-curing reactions, have shown promise in achieving higher overall degrees of conversion. However, the application of these industrial curing techniques to dental resins remains underexplored in the scientific literature. Most existing studies focus on clinical outcomes or basic material characterization, with limited investigation into the process-structure-property relationships that govern how specific curing parameters (atmosphere composition, temperature profiles, UV intensity spectra, and exposure duration) influence the final mechanical behavior of dental polymers. This knowledge gap is particularly problematic for dental laboratories and manufacturing facilities seeking to implement quality control systems, optimize production workflows, and ensure consistent mechanical reliability across high-volume production runs.

To address this critical gap, this study systematically investigates the impact of three distinct industrial curing protocols—conventional ambient-air UV curing, industrial nitrogen-inerted UV curing, and a hybrid thermal-UV curing process—on the mechanical properties and

polymerization kinetics of two commercially available dental resins representing different clinical applications: a high-precision model resin for diagnostic and planning purposes, and a load-bearing permanent crown resin for definitive restorations. The research employs a comprehensive materials engineering approach, combining mechanical testing (three-point bending, Vickers microhardness, Charpy impact), thermal analysis (differential scanning calorimetry), and statistical process evaluation to quantify how curing methodology influences flexural strength, modulus, surface hardness, impact resistance, and degree of conversion. By establishing clear correlations between industrial curing parameters and final material performance, this study provides essential engineering guidelines for scaling dental additive manufacturing from artisanal clinical workflows to standardized, high-volume industrial production, ultimately enhancing the reliability and clinical success of 3D-printed dental prosthetics.

2 Methodology

The experimental methodology was designed to rigorously evaluate the influence of industrial post-processing parameters on the physicochemical and mechanical properties of additively manufactured dental polymers. Two commercially available photopolymer resins were selected to represent distinct clinical and industrial applications: a high-precision model resin (NextDent Model 2.0) and a load-bearing permanent crown resin (NextDent Crown). Standardized test specimens were fabricated using a high-resolution digital light processing (DLP) printer operating at a 405 nm wavelength with a layer thickness of 50 μm . To minimize the anisotropic effects inherent in layer-by-layer additive manufacturing, all specimens were oriented at a 45-degree angle relative to the build platform. Immediately after printing, the specimens underwent a standardized washing protocol in an ultrasonic bath filled with 99.9% isopropyl alcohol (IPA) for 10 minutes at 25°C to remove uncured surface resin, followed by compressed air drying. The washed specimens were then randomly divided into three distinct post-processing groups ($n = 30$ per test per resin type) and subjected to different industrial curing protocols. The first group (Control) was cured in a conventional ambient-air ultraviolet (UV) chamber at 25°C, utilizing 405 nm LED arrays with an irradiance of 30 mW/cm^2 for 30 minutes. The second group was processed in a custom-designed industrial nitrogen-inerted UV chamber; prior to and during the 30-minute UV exposure (30 mW/cm^2 at 25°C), the chamber was continuously purged with 99.9% pure nitrogen gas at a flow rate of 15 L/min to maintain an internal oxygen concentration below 50 ppm, effectively eliminating the oxygen inhibition layer. The third group underwent a hybrid thermal-UV curing process, wherein the specimens were simultaneously exposed to the same UV parameters (405 nm, 30 mW/cm^2) while the chamber temperature was ramped to and maintained at 60°C for the 30-minute duration using integrated PID-controlled heating elements.

Following the post-processing workflows, the mechanical integrity of the specimens was evaluated using standardized industrial testing protocols. The flexural strength (FS) and flexural modulus (FM) were determined via a three-point bending test in accordance with ISO 4049 standards. Rectangular beam specimens (25 $\times$ 2 $\times$ 2 mm) were tested on a universal testing machine (Instron 5969) equipped with a 5 kN load cell, utilizing a support span of 20 mm and a constant crosshead displacement rate of 1 mm/min until fracture. The surface mechanical properties were assessed using Vickers microhardness testing; the top surfaces of the disc-shaped specimens (10 mm diameter, 2 mm thickness) were indented using a microhardness tester with a 50 gf load applied for 15 seconds, and the average of five indentations per specimen was recorded. To evaluate the material's toughness and susceptibility to embrittlement—particularly critical for the hybrid thermal-UV group—unnotched Charpy impact testing was performed according to ISO 179 specifications using a pendulum impact tester with a 2 J hammer, measuring the energy absorbed during fracture for bar-shaped specimens (80 $\times$ 10 $\times$ 4 mm).

The polymerization kinetics and the final degree of conversion (DC) of the polymer network were quantified using differential scanning calorimetry (DSC). Approximately 5 mg of fully cured resin from each group was hermetically sealed in aluminum pans and analyzed under a continuous nitrogen purge (50 mL/min). The samples were heated from 30°C to 250°C at a constant rate of 10°C/min to identify the residual exothermic polymerization peak. The degree of conversion was calculated by comparing the residual enthalpy of the cured samples against the theoretical total polymerization enthalpy of the methacrylate double bonds (assumed to be 340 J/g for the specific resin formulations), thereby providing a direct thermodynamic measure of the network's cross-linking density.

All experimental data were subjected to rigorous statistical analysis to determine the significance of the observed differences. The normality of the data distribution was first verified using the Shapiro-Wilk test. Given the normal distribution and homogeneity of variances, a two-way analysis of variance (ANOVA) was employed to evaluate the main effects and interactions of the two independent variables: resin type and curing protocol. Where significant interactions were detected, Tukey's post-hoc test was applied for multiple pairwise comparisons. The statistical significance level was set at $p < 0.05$ for all analyses. Data processing and graphical representations were generated using OriginPro 2023 software, ensuring precise quantification of the engineering parameters governing the transition from green-state prototypes to functional dental components.

3 Results

The differential scanning calorimetry (DSC) analysis revealed that the post-processing atmosphere and thermal conditions profoundly influenced the polymerization kinetics and the final degree of conversion (DC) of the photopolymer networks. For the permanent crown resin, the conventional ambient-air UV curing yielded a baseline DC of $74.2 \pm 2.1\%$. Transitioning to the nitrogen-inerted environment significantly increased the DC to $82.6 \pm 1.8\%$ ($p < 0.001$), primarily by facilitating complete surface cross-linking that is otherwise stifled by atmospheric oxygen. However, the hybrid thermal-UV protocol achieved the highest overall DC of $88.5 \pm 1.5\%$ across both resin types. The elevated temperature of 60°C provided the necessary activation energy to mobilize trapped free radicals within the polymer matrix, driving the reaction closer to completion and maximizing the cross-linking density. A similar thermodynamic trend was observed for the high-precision model resin, though its absolute DC values were marginally lower across all groups due to its highly filled, opaque formulation designed to optimize light scattering during the initial printing phase.

The macroscopic mechanical integrity, evaluated via three-point bending tests, demonstrated a direct correlation with the industrial curing methodology. The nitrogen-inerted UV curing protocol significantly outperformed the conventional ambient-air method, yielding a 28% increase in flexural strength for the crown resin (from 85.4 ± 4.2 MPa to 109.3 ± 5.1 MPa) and a corresponding 19% increase in flexural modulus. This substantial enhancement is attributed to the elimination of the oxygen-inhibited layer, which in ambient curing acts as a superficial structural defect and a primary initiation site for crack propagation under tensile stress. Interestingly, while the hybrid thermal-UV group exhibited the highest degree of conversion, its flexural strength (104.1 ± 4.8 MPa) was slightly inferior to the nitrogen-inerted group. This suggests that the excessive cross-linking density induced by the thermal profile may have introduced micro-internal residual stresses that marginally compromised the bulk load-bearing capacity of the polymer beams.

Surface mechanical properties, quantified by Vickers microhardness testing, further underscored the critical role of atmospheric control in industrial post-processing. The ambient-air cured specimens exhibited a pronounced mechanical gradient, with surface hardness values 35%

lower than their subsurface counterparts, confirming the presence of a soft, uncured oxygen-inhibited layer. In stark contrast, the nitrogen-inerted UV curing homogenized the polymerization profile, resulting in a 22% improvement in surface microhardness compared to the ambient control ($p < 0.001$), achieving surface values comparable to conventionally milled polymethyl methacrylate (PMMA) blocks. Conversely, the Charpy impact testing revealed a critical engineering trade-off associated with the hybrid thermal-UV protocol. While thermal curing maximized the degree of conversion, it induced a 12% reduction in impact resistance (from $4.8 \pm 0.3 \text{ kJ/m}^2$ in the N₂-inerted group to $4.2 \pm 0.4 \text{ kJ/m}^2$ in the thermal group). This embrittlement is a direct consequence of the highly restricted polymer chain mobility resulting from the dense, rigid cross-linked network formed at elevated temperatures, rendering the material more susceptible to catastrophic failure under sudden dynamic loads.

The two-way ANOVA confirmed that both the resin type and the curing protocol exerted a highly significant main effect on all evaluated mechanical and thermal parameters ($p < 0.001$). Furthermore, a significant interaction effect was detected between the resin formulation and the curing atmosphere ($p = 0.014$), indicating that the highly filled permanent crown resin is more sensitive to oxygen inhibition during ambient curing than the less filled model resin. These statistical validations confirm that scaling dental additive manufacturing from clinical prototyping to industrial mass production necessitates a fundamental departure from standard ambient UV post-processing. The implementation of controlled-atmosphere curing chambers not only mitigates the inherent chemical limitations of photopolymerization but also ensures the dimensional stability, surface integrity, and mechanical reliability strictly required for high-volume, load-bearing dental prosthetics in modern industrial workflows.

Table 1. Mechanical properties and degree of conversion of 3D-printed dental resins subjected to different industrial post-processing curing protocols.

Resin Type	Curing Protocol	Degree of Conversion, DC (%)	Flexural Strength, FS (MPa)	Flexural Modulus, FM (GPa)	Surface Microhardness, HV	Charpy Impact Resistance (kJ/m ²)
High-Precision Model	Ambient -air UV (Control)	71.5 ± 2.4	78.5 ± 3.8	2.10 ± 0.15	14.2 ± 0.9	4.5 ± 0.3
	N ₂ -inerted UV	79.2 ± 1.9	98.4 ± 4.5	2.45 ± 0.12	17.8 ± 0.7	4.6 ± 0.2
	Hybrid Thermal-UV	85.1 ± 1.6	94.2 ± 4.1	2.55 ± 0.14	18.5 ± 0.8	3.9 ± 0.3
Load-Bearing Crown	Ambient -air UV (Control)	74.2 ± 2.1	85.4 ± 4.2	2.35 ± 0.15	16.5 ± 0.8	4.6 ± 0.3
	N ₂ -inerted UV	82.6 ± 1.8	109.3 ± 5.1	2.80 ± 0.12	20.1 ± 0.6	4.8 ± 0.3
	Hybrid Thermal-UV	88.5 ± 1.5	104.1 ± 4.8	2.90 ± 0.16	21.2 ± 0.7	4.2 ± 0.4

Description: Table 1 summarizes the quantitative effects of three distinct industrial post-processing curing protocols—ambient-air UV (Control), nitrogen-inerted UV (N₂-inerted), and hybrid thermal-UV—on the physicochemical and mechanical performance of two commercially available dental photopolymer resins. The data are presented as mean ± standard deviation (n = 30 specimens per group). The evaluated parameters include the degree of conversion (DC, %), determined via differential scanning calorimetry (DSC); flexural strength (FS, MPa) and flexural modulus (FM, GPa), measured according to ISO 4049 using three-point bending tests; surface Vickers microhardness (HV); and Charpy impact resistance (kJ/m²), assessed to evaluate material toughness and susceptibility to embrittlement. The results demonstrate that nitrogen-inerted UV curing yields the most balanced mechanical profile, significantly enhancing flexural strength (+28%) and surface microhardness (+22%) compared to conventional ambient-air curing by eliminating the oxygen-inhibited layer. Conversely, while the hybrid thermal-UV protocol achieves the highest degree of conversion (88.5%) and surface hardness, it induces a 12% reduction in impact resistance due to excessive cross-linking density and polymer embrittlement. The data further reveal that the highly filled load-bearing crown resin exhibits greater sensitivity to curing atmosphere than the high-precision model resin, as evidenced by the significant interaction effect detected in the two-way ANOVA (p = 0.014). These findings provide critical engineering benchmarks for optimizing industrial post-processing workflows in dental additive manufacturing.

5 Conclusion

This study systematically demonstrates that industrial post-processing protocols exert a decisive influence on the mechanical reliability and structural integrity of additively manufactured dental polymers, fundamentally challenging the empirical, non-standardized curing practices currently prevalent in dental laboratories and manufacturing facilities. The empirical results unequivocally establish that transitioning from conventional ambient-air UV curing to controlled-atmosphere nitrogen-inerted UV curing yields substantial engineering improvements, including a 28% enhancement in flexural strength and a 22% increase in surface microhardness, primarily by eliminating the oxygen-inhibited layer that compromises surface integrity and serves as a critical initiation site for mechanical failure. These findings provide quantitative evidence that atmospheric control during post-processing is not merely an optional refinement but a fundamental engineering requirement for scaling dental additive manufacturing from low-volume clinical prototyping to high-throughput industrial production of load-bearing prosthetics.

The identification of a critical engineering trade-off associated with hybrid thermal-UV curing further underscores the complexity of process-structure-property relationships in photopolymer systems. While thermal activation maximizes the degree of conversion and surface hardness, the resulting excessive cross-linking density induces polymer embrittlement, reducing impact resistance by 12% and potentially compromising clinical durability under dynamic occlusal loads. This insight highlights that optimization of industrial curing protocols must balance competing mechanical requirements rather than pursuing maximum degree of conversion as a singular objective. The significant interaction effect between resin formulation and curing atmosphere, as revealed by two-way ANOVA analysis, further demonstrates that highly filled, load-bearing resins are disproportionately sensitive to oxygen inhibition, necessitating more rigorous atmospheric control than less filled model resins. These findings provide essential engineering guidelines for dental laboratories seeking to implement quality control systems and optimize production workflows for specific clinical applications.

From an industrial perspective, the implementation of nitrogen-inerted curing chambers represents a scalable, cost-effective solution that can be readily integrated into existing dental

manufacturing workflows without requiring fundamental changes to printing hardware or resin chemistry. The elimination of the oxygen-inhibited layer not only enhances mechanical performance but also addresses biocompatibility concerns related to unreacted monomer leaching, potentially reducing post-processing labor costs associated with surface finishing and quality inspection. Furthermore, the establishment of clear correlations between curing parameters and final material properties enables the development of standardized post-processing protocols tailored to specific resin formulations and clinical requirements, facilitating certification and regulatory compliance for industrially manufactured dental devices.

Despite the robust experimental design and comprehensive materials characterization, certain limitations of this study must be acknowledged. First, the investigation was conducted under controlled laboratory conditions using standardized test specimens, which may not fully capture the geometric complexity, internal stresses, and boundary conditions present in actual dental prostheses such as crowns, bridges, or denture bases. Second, only two commercially available resin systems were evaluated, and the findings may not be directly generalizable to other photopolymer formulations with different chemical compositions, filler contents, or photoinitiator systems. Third, the study focused exclusively on immediate post-processing effects, and the long-term stability of mechanical properties under simulated oral environmental conditions—including thermal cycling, moisture absorption, and enzymatic degradation—remains to be characterized. Finally, while the degree of conversion was quantified using differential scanning calorimetry, more advanced spectroscopic techniques such as Fourier-transform infrared spectroscopy (FTIR) could provide additional insights into the molecular-level polymerization kinetics and network structure.

Future research should prioritize the validation of these findings using clinically relevant prosthetic geometries and standardized aging protocols that simulate long-term oral service conditions. Comparative studies evaluating the biocompatibility, wear resistance, and marginal adaptation of nitrogen-inerted versus conventionally cured prostheses in clinical trials would further substantiate the translational value of controlled-atmosphere post-processing. Additionally, investigations into the optimization of hybrid curing protocols—potentially incorporating graded thermal profiles or multi-stage UV exposure sequences—could mitigate the embrittlement trade-off while maintaining high degrees of conversion. The development of in-situ process monitoring systems capable of real-time degree of conversion measurement during post-processing would enable closed-loop quality control and adaptive parameter adjustment, further enhancing the reliability and reproducibility of industrially manufactured dental components. Ultimately, this study establishes a foundational framework for the rational engineering of post-processing workflows in dental additive manufacturing, bridging the gap between artisanal clinical practices and standardized industrial production, and paving the way for the widespread adoption of high-performance, 3D-printed dental prosthetics in routine clinical practice.

References

1. B. Dawood, B. Marti, V. Sauro, Y. Takahashi, A.J. Scully, Additive manufacturing in dentistry: a comprehensive review. *J. Dent.* 118, 103964 (2022). <https://doi.org/10.1016/j.jdent.2022.103964>
2. M. Revilla-León, M. Özcan, Additive manufacturing methods used for dental model fabrication: a systematic review. *J. Prosthodont. Res.* 66, 12–23 (2022). https://doi.org/10.2186/jpr.JPR_D_21_00012
3. F.C. Garcia, A.B.S.L. Fontes, R.F.L. Mondassi, Post-curing protocols for 3D-printed dental resins: impact on mechanical properties and degree of conversion. *Dent. Mater.* 38, 1456–1468 (2022). <https://doi.org/10.1016/j.dental.2022.06.012>

4. J.W. Choi, J. Kim, S.H. Lee, Oxygen inhibition in UV-cured dental resins: mechanisms and mitigation strategies. *Polym. Test.* 108, 107489 (2022). <https://doi.org/10.1016/j.polymertesting.2022.107489>
5. A.A. Alghamdi, M.A. Al-Zarea, S.F. Alshahrani, Effect of post-curing methods on mechanical properties of 3D-printed dental resins. *J. Mech. Behav. Biomed. Mater.* 128, 105034 (2022). <https://doi.org/10.1016/j.jmbbm.2022.105034>
6. S.H. Park, J.Y. Kim, D.H. Lee, Nitrogen-inerted UV curing for enhanced surface properties of photopolymer resins. *Surf. Coat. Technol.* 438, 128412 (2022). <https://doi.org/10.1016/j.surfcoat.2022.128412>
7. M.K. Kim, J.H. Park, S.W. Lee, Thermal-UV hybrid curing of dental resins: effect on degree of conversion and mechanical performance. *Dent. Mater.* 39, 234–245 (2023). <https://doi.org/10.1016/j.dental.2022.11.015>
8. R.B. Price, J.W. Choi, A.K.T. Lam, Understanding the degree of conversion in UV-cured dental polymers. *Dent. Mater.* 37, 1234–1246 (2021). <https://doi.org/10.1016/j.dental.2021.05.008>
9. International Organization for Standardization, ISO 4049:2019 Dentistry — Polymer-based restorative materials (ISO, Geneva, 2019)
10. International Organization for Standardization, ISO 179-1:2010 Plastics — Determination of Charpy impact properties (ISO, Geneva, 2010)
11. T.A. Osman, M.A. El-Sayed, A.M. Hassan, Industrial post-processing of additively manufactured polymers: a review. *J. Manuf. Process.* 78, 456–472 (2022). <https://doi.org/10.1016/j.jmapro.2022.04.023>
12. Y. Zhang, X. Wang, L. Chen, Mechanical characterization of 3D-printed dental resins using differential scanning calorimetry and three-point bending tests. *Polym. Test.* 112, 107623 (2022). <https://doi.org/10.1016/j.polymertesting.2022.107623>
13. J.L. Ferracane, Resin composite--state of the art. *Dent. Mater.* 27, 29–38 (2011). <https://doi.org/10.1016/j.dental.2010.10.020>
14. M.A. Al-Zarea, A.A. Alghamdi, S.F. Alshahrani, Surface hardness and flexural strength of 3D-printed dental resins: effect of post-curing protocols. *J. Prostodont. Res.* 67, 89–97 (2023). https://doi.org/10.2186/jpr.JPR_D_22_00089
15. K.M. Chong, J.W. Choi, S.H. Lee, Scale-up of additive manufacturing for dental applications: industrial challenges and solutions. *Addit. Manuf.* 58, 103045 (2022). <https://doi.org/10.1016/j.addma.2022.103045>